Audio System Isolation Amplifier – General data

1. General

The NAT 247 audio isolation amplifier is used to mix warning audio alerts from various aircraft systems.

2. Description

A. Audio Isolation Amplifier (NAT 247)

The audio isolation amplifier is used as a mixing amplifier to integrate various aircraft audio signals into the audio control panel. The audio isolation amplifier is mounted on the port side of the middle forward avionics shelf.

28 Volt DC power is protected by a 5 amp PILOT AUDIO (3E) (Left Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The NAT 247 audio isolation amplifier interfaces to the following aircraft systems:

1) Audio Control Panel (refer to 23-53-00, Audio Control Panel – General Data).

2) Traffic Alert and Collision Avoidance System (TCAS I and TCAS II) (refer to 34-43-00, Traffic Alert and Collision Avoidance System I (TCAS I) – General Data and 34-43-01, Traffic Alert and Collision Avoidance System II (TCAS II) (Mod S.O.O. 6219) – General Data).

3) Enhanced Ground Proximity Warning System (EGPWS) (TAWS) (refer to 34-44-00, Enhanced Ground Proximity Warning System (TAWS) (Mod 6/2049) – General Data).

3. Adjustment/Test

A. Function Test Audio Isolation Amplifier (NAT 247)

1) Function Test Audio Isolation Amplifier

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that PILOT AUDIO (3E) circuit breaker is closed.

d) Connect a headset to the pilot, and/or co-pilot position.

e) In the APEX system sensors page, initiate a TCAS self test, (if installed).

f) Verify TCAS system test aural alert is audible in the headset.

g) In the APEX system sensors page, initiate a self test.

h) Verify TAWS system test aural alert is audible in the headset.

i) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT/REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal and Installation of the Audio Isolation Amplifier (NAT 247)

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the PILOT AUDIO (3E) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Gain access to the forward avionics shelves through the forward baggage compartment.

d) Disconnect the electrical connector on the front of the unit.

e) Remove the screws (4) securing the unit to the forward avionics shelf.

f) Remove the unit from aircraft.

Note Stow connectors as required.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the PILOT AUDIO (3E) circuit breaker on the avionics circuit breaker panel is opened and fitted with safety clip.

c) Align the unit mounting holes to the forward avionics shelf.

d) Secure the unit to the avionics shelf with mounting screws (4).

e) Reconnect connector to the unit. Ensure connections are secure.

f) Remove safety clip and close PILOT AUDIO (3E) circuit breaker on the avionics circuit breaker panel.

g) Test the Audio Isolation Amplifier as per Para 3.A.

5. Troubleshooting

A. Audio Isolation Amplifier (NAT 247)

1) If the Audio Isolation Amplifier (NAT 247) is deemed inoperative check the following:

a) Ensure PILOT AUDIO (3E) circuit breaker is closed.

b) Visually inspect audio isolation amplifier connector for damaged wires or contacts.

c) Ensure that the audio isolation amplifier connector is properly mated and secure.

d) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

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